DISPATCH
@IETF127

LinkID: questions and answers

Backup for draft-linkgenetic-linkid-uri-01

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FAQ: what LinkID is

Isn’t this just a redirect service or a PURL?

- No. A LinkID is not bound to a resolver’s domain; any conformant resolver can answer.
- Each answer comes from a signed record with lifecycle status (active, revoked, tombstoned).
- Withdrawn identifiers return 410 and are never reassigned.

Why a new URI scheme and not a URN?

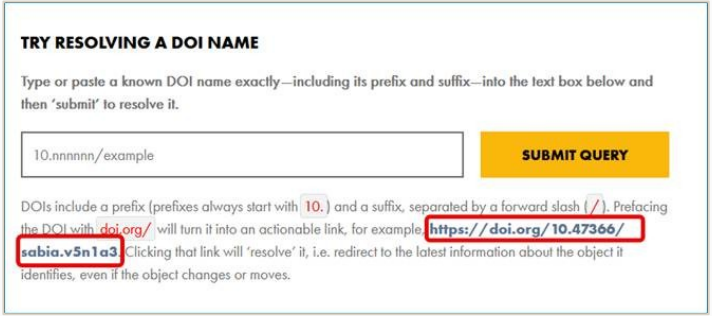
- urn:uuid identifies a thing but defines no resolution, lifecycle or HTTPS binding.
- The resolution and lifecycle model is the substance; the scheme name is secondary.
- We are open to a URN namespace if the community prefers it.

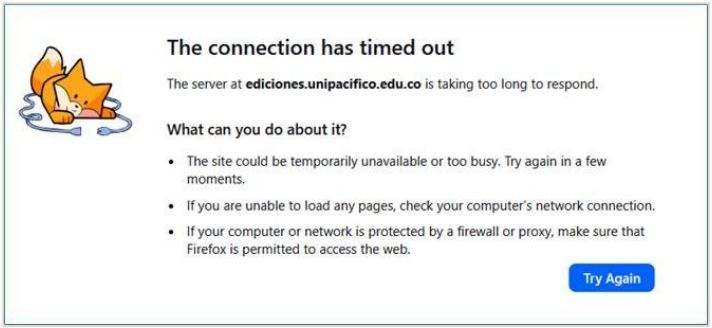
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FAQ: relation to DOI

Why not use DOI?

- DOI guarantees the identifier; keeping the target reachable is the registrant's duty.
- doi.org's own example DOI resolved to an unreachable host in May 2026.
- 4–15% of sampled DOIs end in an error · Klein, Balakireva 2020
- 708,282 Crossref DOIs deleted 2017–2021 · Kikkawa et al. 2022
- Complementary, not competing: a DOI is a valid LinkID target.





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FAQ: scale

Does every resolver need to know every LinkID?

- No. Records are signed and portable; resolvers fetch, verify and cache on demand.
- The same pattern as DNS: delegation, caching, many independent operators.
- Today: the linkid.io resolver, delivered through a CDN with 450+ edge locations.
- Independent operators: none active yet; an open-source failover resolver is prepared on Wikimedia Toolforge, not yet in service. CDN reach is not the same as independent resolvers.

How do independent resolvers find each other?

- Today the primary publishes a discovery chain of mirrors.
- Discovery without a central list is an open issue; input is welcome.

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FAQ: security

Why should I trust the resolver?

- You don’t have to: portable resolution records are Ed25519-signed; public keys are published as JWKS.
- Mirrors verify the record offline and fail closed: no trusted key, no answer.
- Platform identifier evidence uses a separate ECDSA P-256 signature; not every redirect is signed.

Can A claim or change B’s LinkID?

- Today: authenticated, audited updates; one platform signing key.
- Direction for -02: keys per Registration Agency; transfers signed by the previous key.
- Key loss and recovery are still open.

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FAQ: independence and governance

What happens if Link Genetic disappears?

- Signed records can be served from a record host or IPFS; the open-source failover resolver is designed to run without linkid.io (prepared on Wikimedia Toolforge, not yet in service).
- Long term, the trust anchor has to move to neutral governance.

Isn’t Link Genetic a single point of control?

- Today, yes, for minting and the signing key.
- Every system has a root of trust; the DNS root is run by 12 organisations in several countries.
- Goal: Registration Agencies with their own keys and a neutral governance body (next slide).

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FAQ: governance model (proposed)

Who governs LinkID in the long term? Not Link Genetic alone.

- **Govern:** a nonprofit LinkID Foundation, independent of Link Genetic, stewards the spec and the linkid scheme and accredits providers. No group holds a board majority; Link Genetic holds one seat.
- **Operate:** multiple accredited service providers run interoperable resolvers; Link Genetic is one of them. An institution can run its own.
- **Steward:** a global community flags broken or drifted targets and proposes fixes; the owner decides, and every change is logged.

Where does this stand today?

- A proposal: the foundation, its board and further providers do not exist yet; Link Genetic is the only operator today.
- Indicative: charter drafted in the open in 2027, first accredited providers H2 2027, handover 2028; depends on partners and funding.
- Open spec and SDKs: github.com/Link-Genetic-Inc/lid; input welcome on the mailing list.

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FAQ: guarantees and privacy

Does a LinkID guarantee that a link never breaks?

- No. A persistent identifier does not guarantee that its target stays reachable.
- Broken targets and content drift are detected and reported; the authority re-points the record.
- When a target problem is detected, LinkID can offer an information or recovery page.
- If no usable target or archive remains, a notice explains what happened; it does not restore lost content.

Does the resolver track who reads what?

- Resolution needs no client identity; any mirror or cache can answer.
- Records restricted to a region are never served by mirrors.

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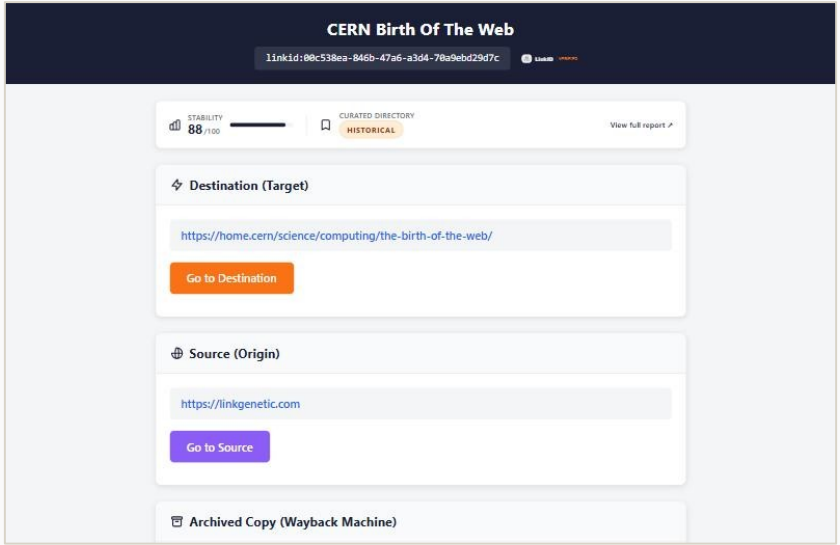
FAQ: what a reader sees

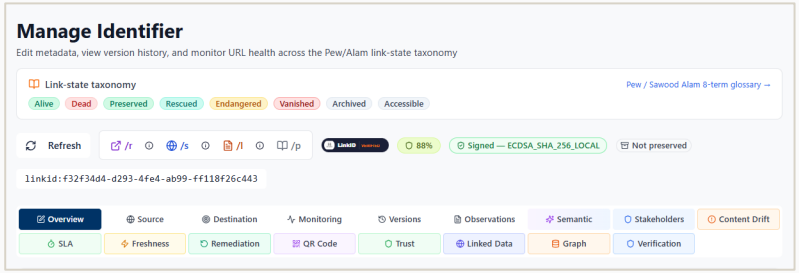
Which resolution routes exist per LinkID?

- /r: default resolution to the current target
- /s: the original source
- /l: landing page with target, source, archive copy and status

And /c and /p? Publication pages, not routes

- /c: a published Collection of LinkIDs
- /p: a content or information page





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FAQ: IPR, interest and venue

What about the patent?

- IPR disclosure #7635; royalty-free commitment for conforming implementations (PATENTS.md); SDKs under Apache-2.0

Who else is interested?

- Discussions with Wikimedia Foundation, Internet Archive and DataCite.
- Public sector: presented at a hackathon of the Swiss Federal Office of Information Technology (BIT), Oct 2026; no commitment yet.
- EU: Commission minutes list Link Genetic GmbH as a member of Task Force 2; participation, not endorsement (next slide).

Why the IETF?

- Permanent URI scheme registration and the resolution protocol belong here; browser integration is at W3C (WICG #238).

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FAQ: EU participation, public record

What does the public record show?

- Link Genetic GmbH is named as a member in the Commission's final minutes:
- Joint plenary of the task forces, 30 Sept 2026
- Task Force 2 plenary, 1 Oct 2026 (section 6)
- Official task force page

Why does it matter for LinkID?

- Task Force 2 works on an interactive second layer behind EU labels, an audio icon and sectoral labelling practice.
- Persistent references can connect a visible label to provenance, status and versioned information.
- Participation is not EU endorsement, certification or adoption of LinkID.

FAQ: continuity beyond a missing target

What happens when a target is gone for good?

- The UUID stays; target addresses and lifecycle information can change without a new identity.
- Readers use /r, /s or /l; Collections (/c) and content or notice pages (/p) are published separately.
- If no usable target or archive remains, creator-approved context explains what is known and what happened.
- A notice preserves context, not lost content. Losing the resolver operator needs independent stewardship.

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LinkID

- Draft: datatracker.ietf.org/doc/draft-linkgenetic-linkid-uri
- Spec and SDKs: github.com/Link-Genetic-Inc/lid
- Resolver: linkid.io · Demo: linkid.linkgenetic.com/demo
- Slides and FAQ: linkid.linkgenetic.com/ietf-127

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